

Structural FEA – Plate Under Load (Stress, Strain & Deformation Study)

Description:

Conducted structural analysis on a flat structural plate under external force to examine total deformation, normal stress, shear stress, and strain distribution. Built the geometry, applied supports and load, and evaluated stress concentration zones and displacement patterns.

Key Highlights:

- Designed CAD model and refined mesh for accuracy
- Applied fixed supports and directional force loading
- Visualized total deformation, shear stress, normal stress, and strain
- Interpreted material behavior under load with stress contour mapping
- Demonstrated understanding of structural stiffness and load paths

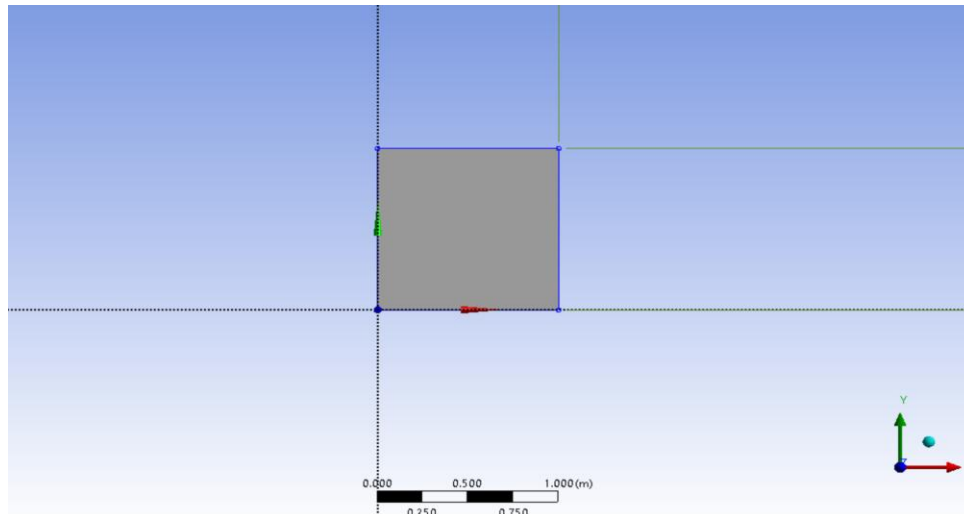
Skills:

Structural Simulation · ANSYS FEA · Stress & Strain Analysis · Mesh Refinement · Engineering Interpretation

Geometry:

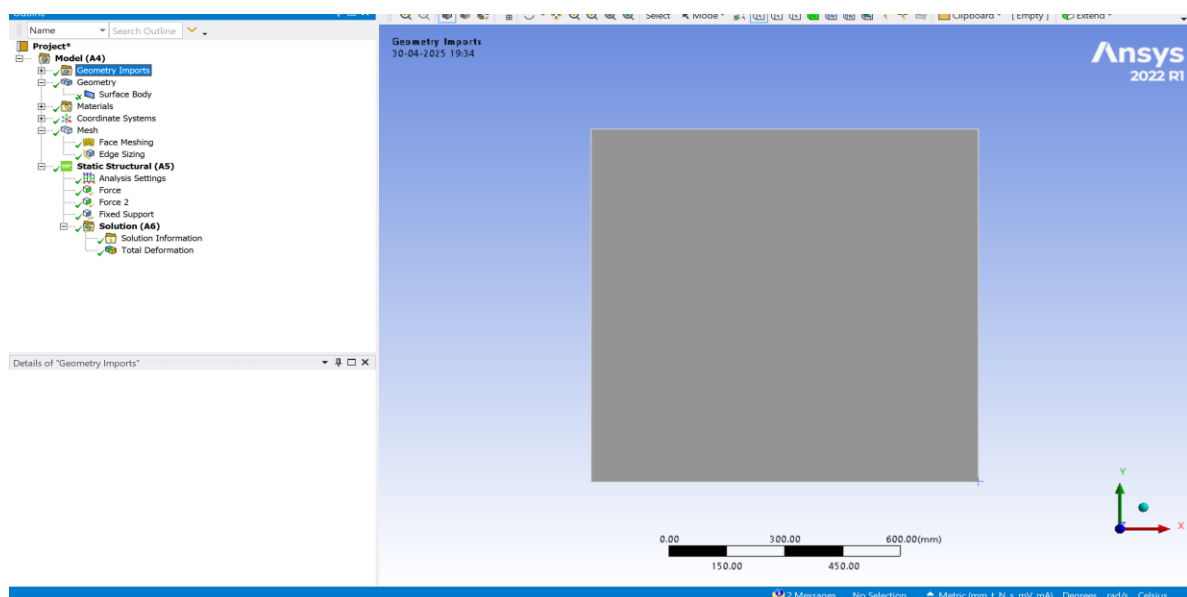
	A	B	C	D	E
1	Property	Value	Unit		
2	 Material Field Variables	 Table			
3	 Density	7850	kg m ⁻³		
4	 Isotropic Secant Coefficient of Thermal Expansior				
6	 Isotropic Elasticity				
7	Derive from	Young's Modulus...			
8	Young's Modulus	10000	MPa		
9	Poisson's Ratio	0.25			

Geometry:

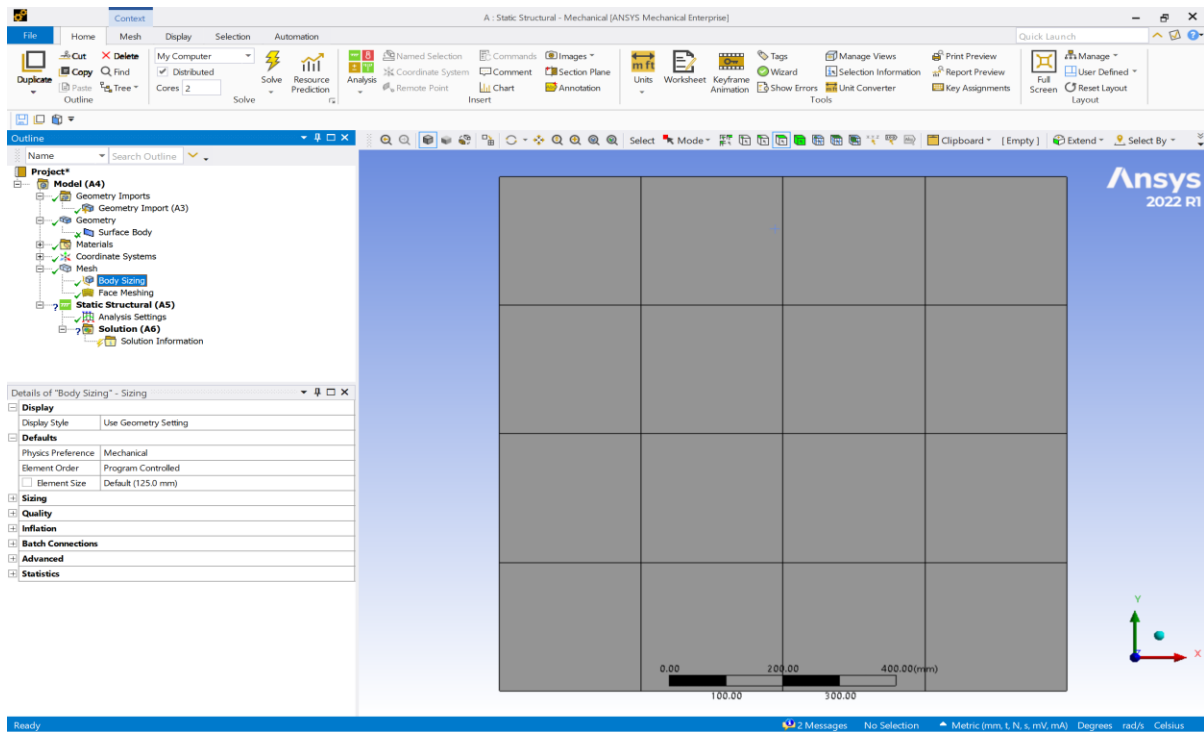


Details View	
Details of Sketch1	
Sketch	Sketch1
Sketch Visibility	Show Sketch
Show Constraints?	No
Dimensions: 2	
H1	1 m
V2	1 m
Edges: 4	
Line	Ln7
Line	Ln8
Line	Ln9
Line	Ln10

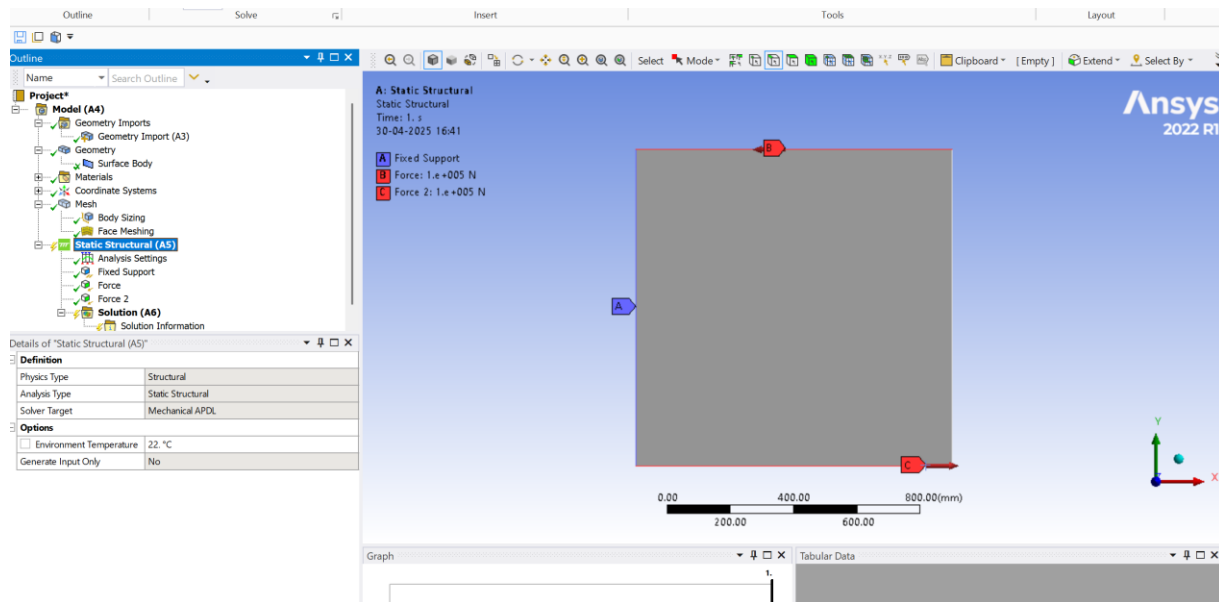
Model:



Mesh:

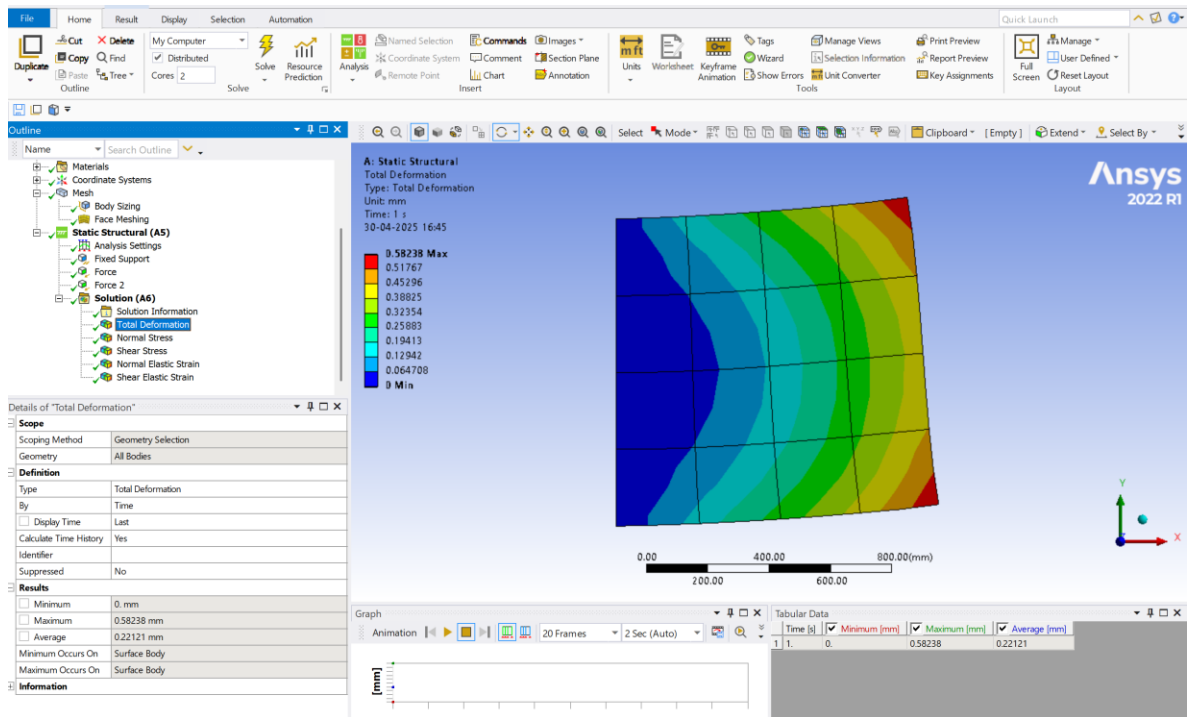


Force:

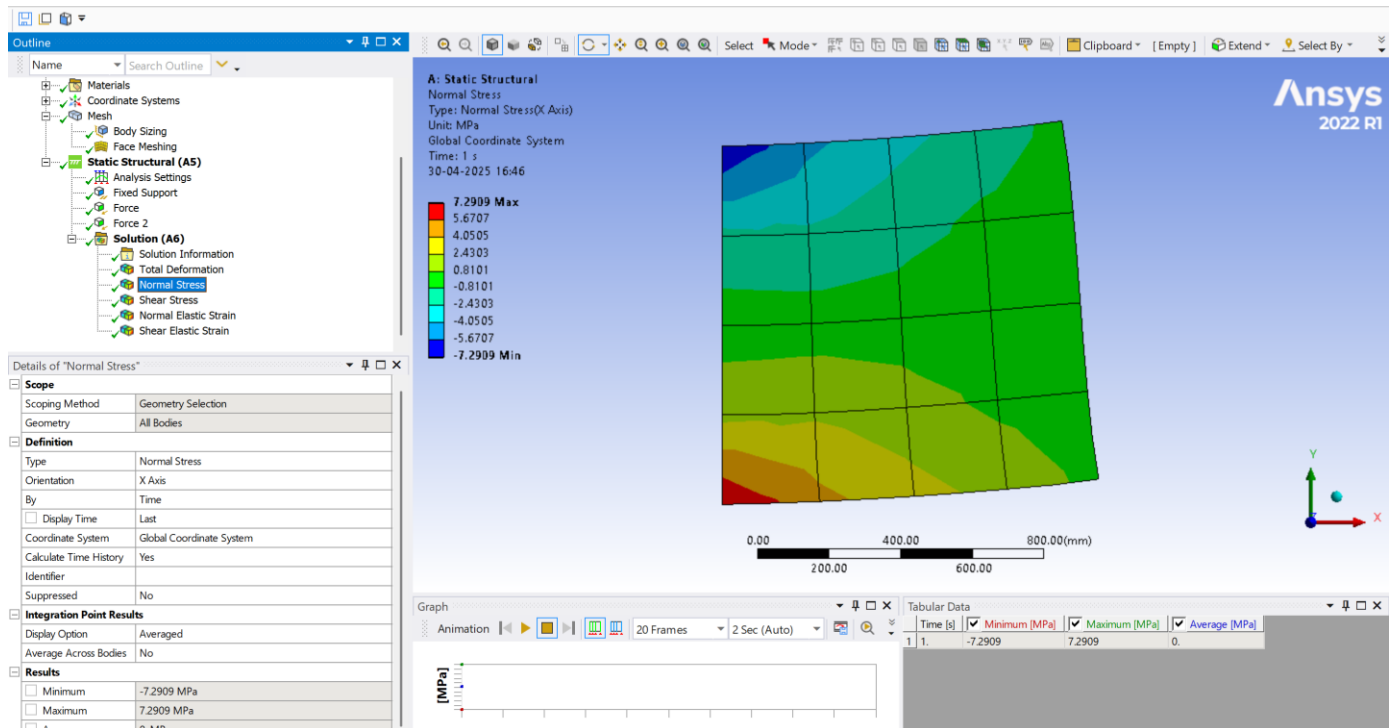


Result:

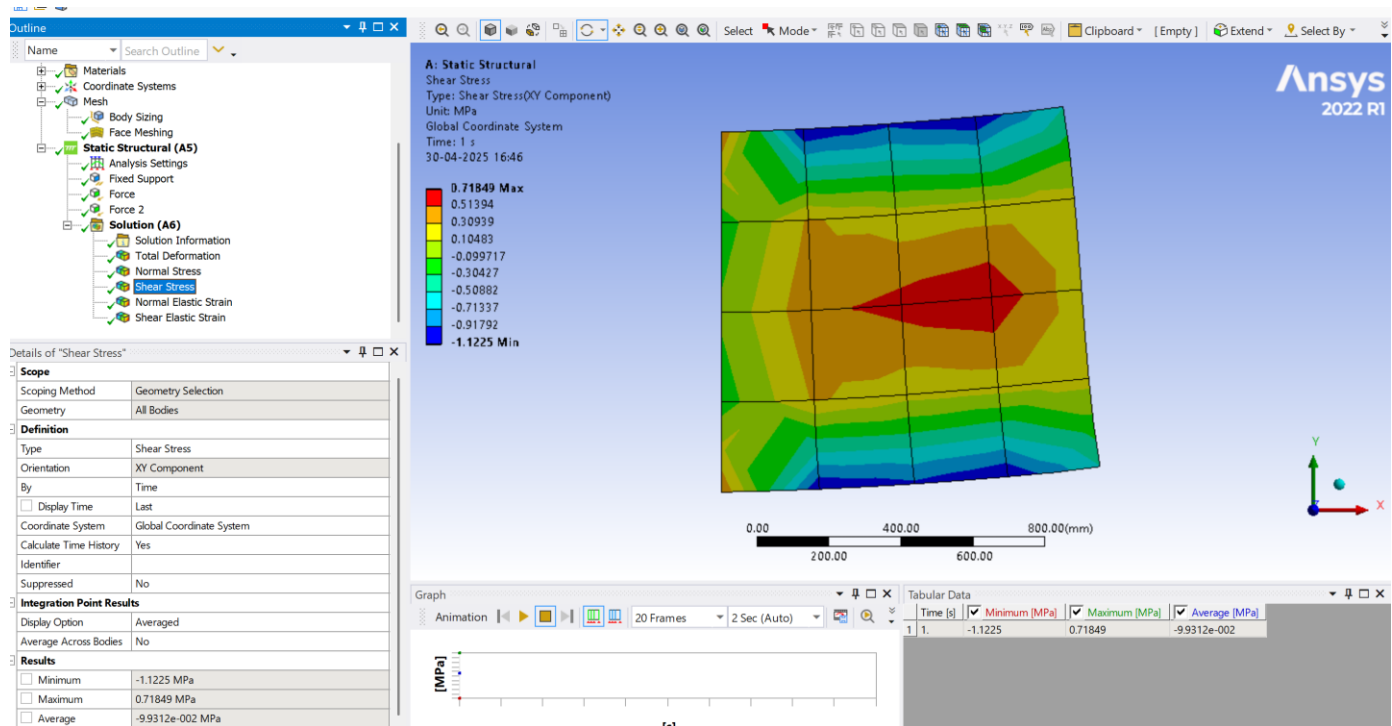
Total Deformation:



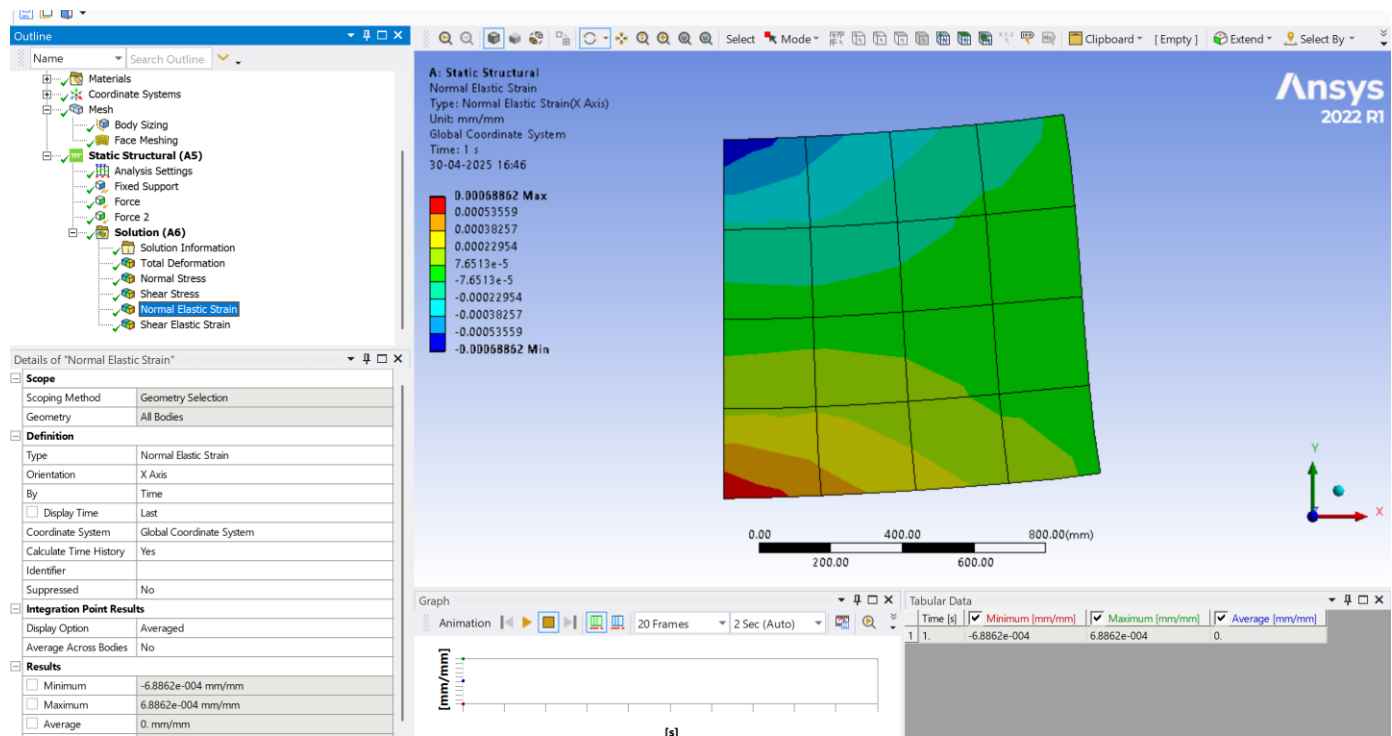
Normal stress:



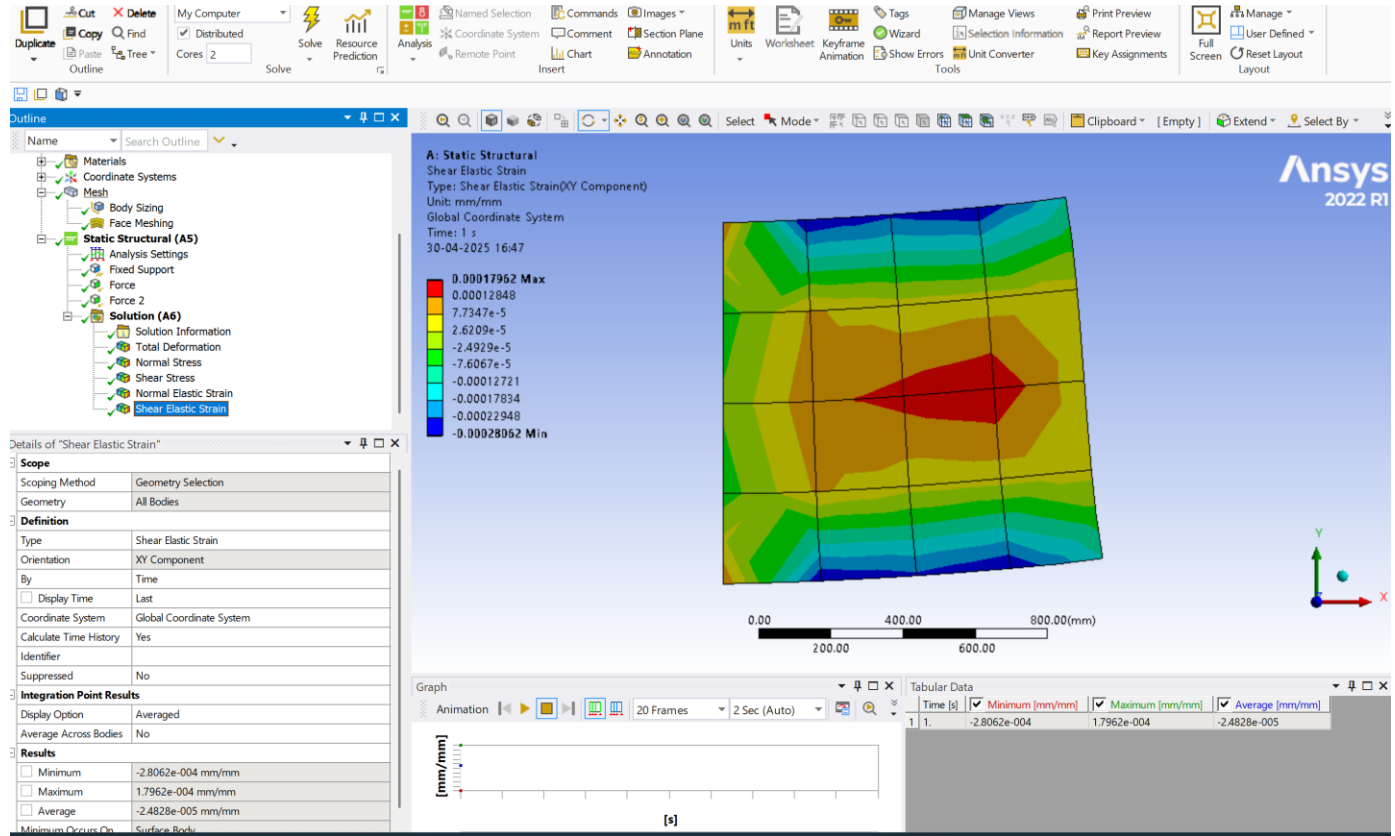
Shear stress:



Normal Strain:

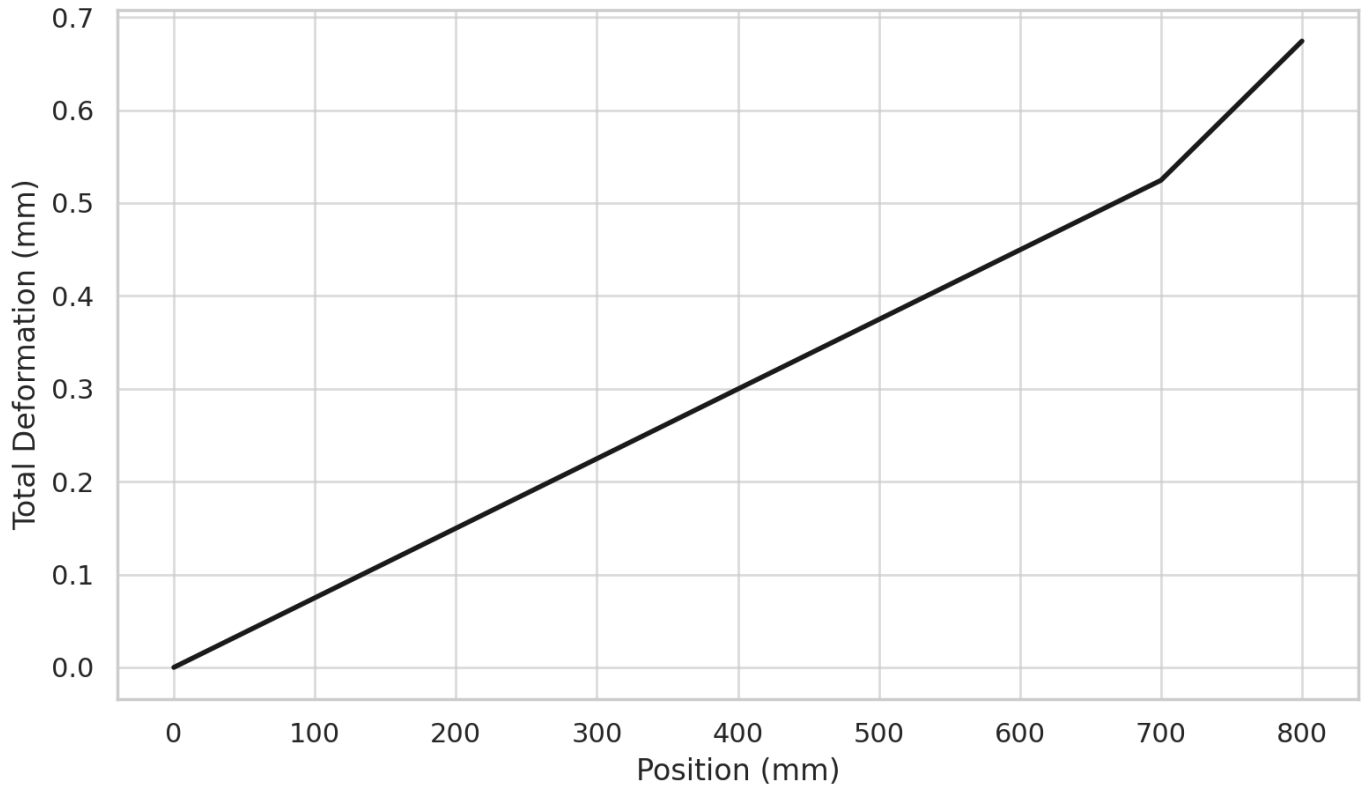


Shear strain:

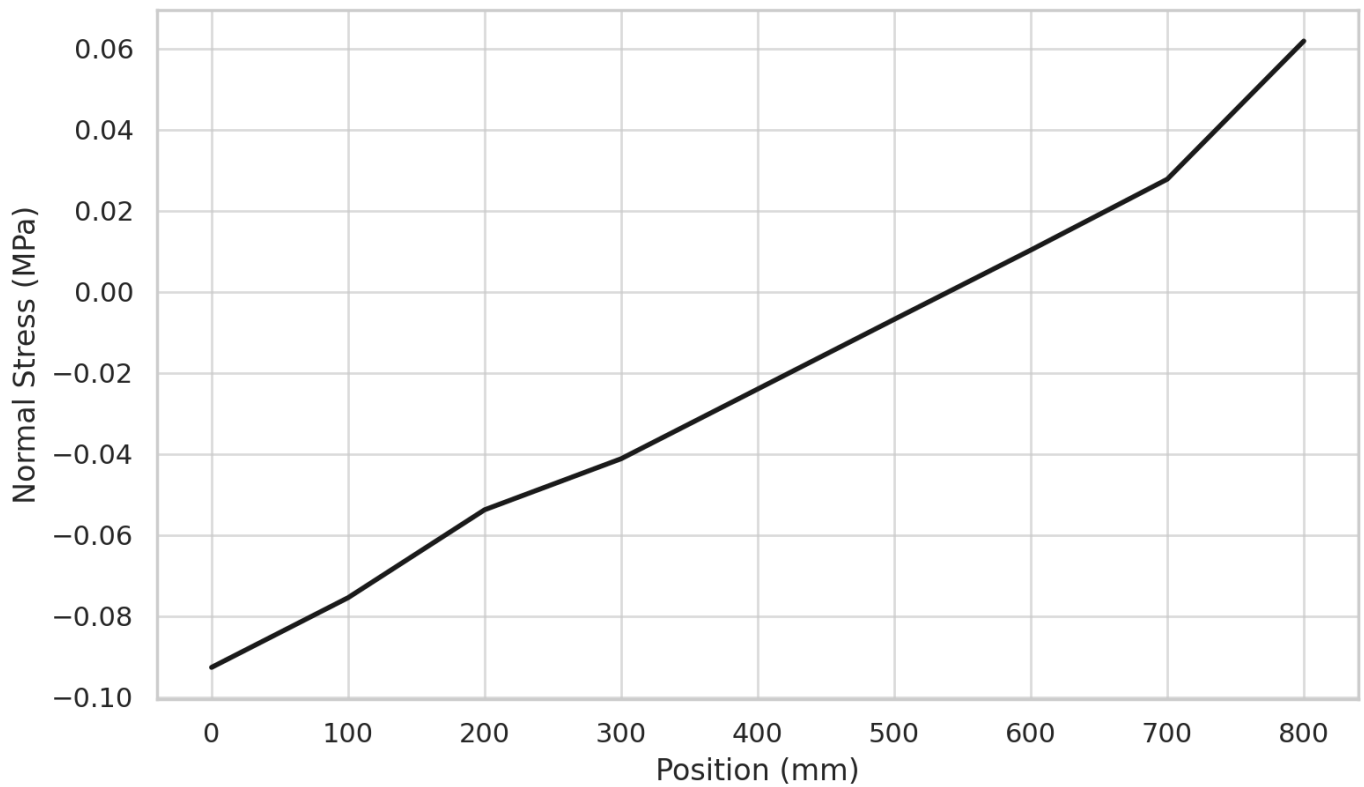


Graphs

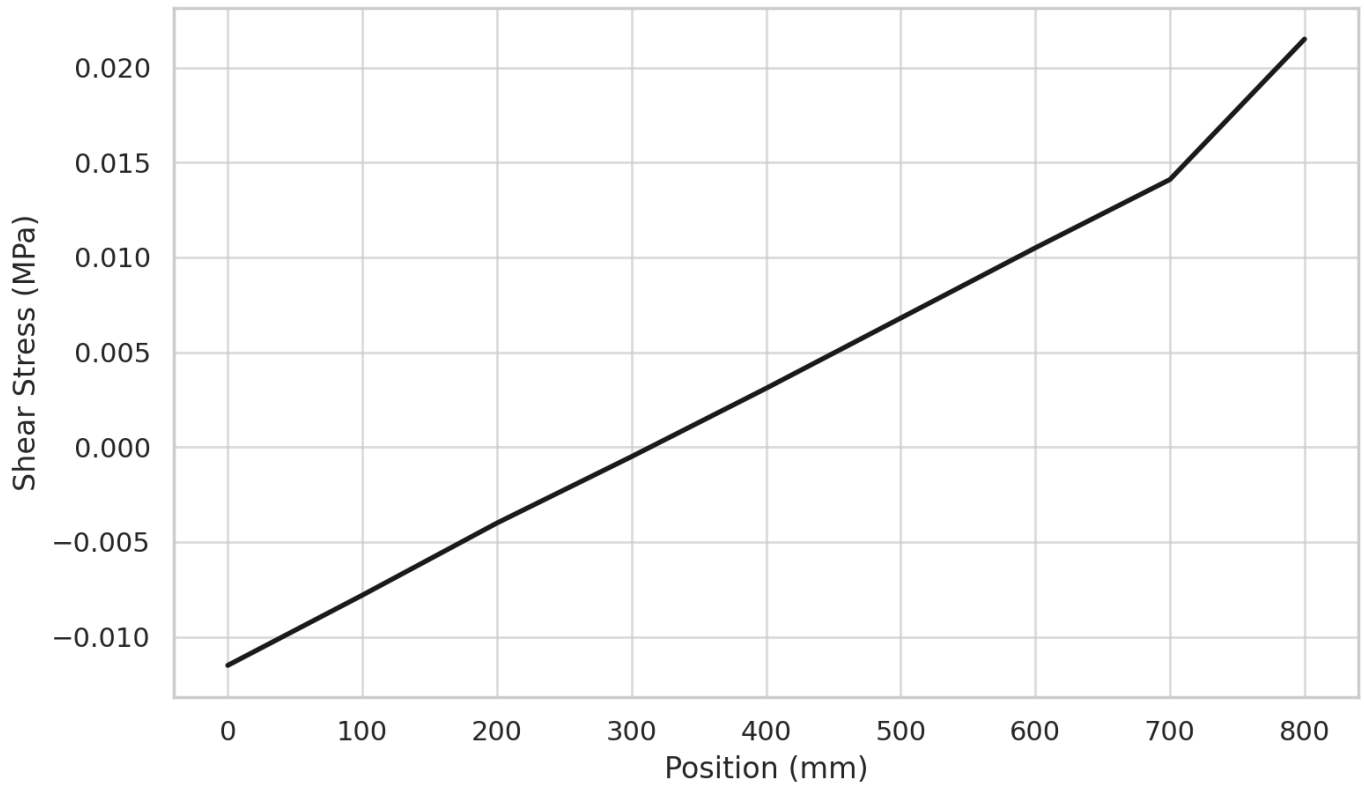
Total Deformation vs Position



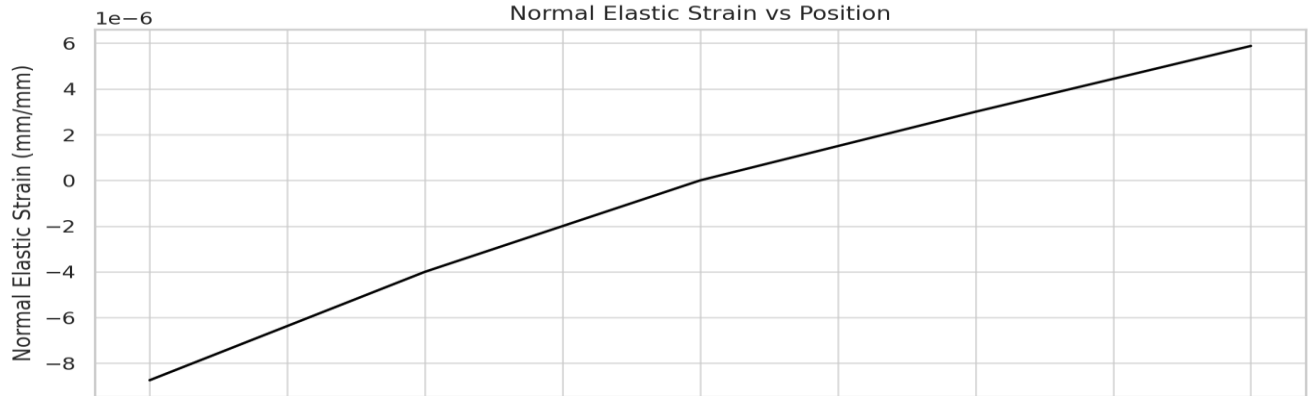
Normal Stress vs Position



Shear Stress vs Position



Normal Elastic Strain vs Position



Shear Elastic Strain vs Position

